The LD PC-V10 Phase Change PC case from LD Cooling with integrated phase change cooling system

is compressed so that sub-zero temperatures can be obtained. Because of this transition between the different states, it's called phase change cooling. Primarily, this system consists of the compressor, evaporator, pump and condenser. The condenser will change the phase of the coolant from a gas to a liquid. As it enters the compressor, the liquid is compressed and routed to the evaporator which is connected to the CPU directly. This is where the compressed liquid evaporates and the coolant in the gas form is sent back to the condenser. You will find single-stage or multi-stage phase cooling where these different phases compress the coolant further. They are called cascades and you can reach really low temperatures with them. The evaporator or vapouriser is present as the CPU block that is directly in contact with the CPU. An advantage of this cooling system is that it can be used for 24/7 cooling. However, they tend to be noisy. There have been different designs for these components, however, the principle has remained the same. Water chillers are commercially available that make use of the same principle. They can be directly connected to your existing liquid cooling system where the coolant passes through it to cool down to freezing temperatures. We do have commercial solutions for phase change cooling with huge single-stage systems placed under the rigs like the LD PC-V10 Phase Change. However, we got to see a prototype from der8auer